

Lab 11: Implementation of SHA-1 Algorithm

Course: Information Security & Cryptography (AI Department, SVNIT Surat)

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Objective

To implement the Secure Hash Algorithm (SHA-1), an iterated hash function that produces a 160-bit message digest. Students will understand message padding, bitwise operations, and the design of cryptographic hash functions.

Background

SHA-1 (Secure Hash Algorithm 1) is a one-way cryptographic hash function that takes an input message of arbitrary length and produces a fixed-size 160-bit output (digest). It processes input blocks of 512 bits and operates on 32-bit words through a sequence of bitwise operations and modular additions.

Algorithm Overview

- 1 Padding the message: Append a single 1 bit, followed by zero bits and message length to make total bits a multiple of 512.
- 2 Initialization: Initialize H0–H4 with fixed constants.
- 3 Processing each 512-bit block: Divide into 16 words W0–W15 and extend to W79 using XOR and left-rotate.
- 4 Main loop: Perform 80 rounds of bitwise and modular operations using constants and round functions.
- 5 Update: Add the temporary hash to the previous hash values.
- 6 Output: Concatenate H0–H4 to form the final 160-bit digest.

Key Observations

- The SHA-1 algorithm converts any input message into a 160-bit digest, ensuring integrity verification.
- It uses bitwise logical operations, circular shifts, and modular additions for non-linearity.
- Small changes in input cause large unpredictable changes in output (avalanche effect).
- SHA-1 is not considered secure for modern cryptographic use due to collision attacks, but is valuable for understanding hashing concepts.

Example Test Cases and Outputs

- 1 Input: 'The quick brown fox jumps over the lazy dog' → Output:
2fd4e1c67a2d28fced849ee1bb76e7391b93eb12
- 2 Input: 'Information Security and Cryptography Lab' → Output:
2fd42e97d9111551984ac20b7e5dfa4432666165

3 Input: " (empty string) → Output: da39a3ee5e6b4b0d3255bfef95601890afd80709

Conclusion

In this lab, the SHA-1 algorithm was successfully implemented in C++. The program correctly performed message padding, block processing, and hash computation. The generated digests matched standard library outputs, confirming the correctness of the implementation. This exercise strengthened understanding of cryptographic hash functions and their internal structure.